Week 45: The visible side

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

Problems and routes

Problem	K–1	2–3	4–5	route
Sort the tickets by hidden color	P1			first
Pairs that show alike but hide different colors		P1		first
Exact hidden-color chances for each clue			P1	first
Red showing: removals that make a tie	P2			first
Blue showing: removals that make a tie	P3			first
The better guess after a red clue		P2		first
The better guess after a blue clue		P3		first
A friend's one-card-each argument			P2	first
One fact about the covered number			P3	first
Add one ticket to the two-ticket chooser	P4			return
Three tickets where red always hides blue	P5			return
Compare the two-ticket chooser with all six tickets		P4	P4	return
Can six rounds tell the choosers apart?			P5	return
Remove one card	P6	P5		return
A two-ticket cup with a fair red clue	P7	P6		return
Every subset with a fair red clue			P6	return
Whole cards cannot make the red clue fair			P7	return

First route. K–1 P1–P3; Grades 2–3 P1–P3; Grades 4–5 P1–P3. **Return visits.** K–1 P4–P7; Grades 2–3 P4–P6; Grades 4–5 P4–P7.

Key, one entry per problem

Sort the tickets by hidden color

K–1 P1 · first route

Idea. Tickets 1, 2 and 4 hide red; 3, 5 and 6 hide blue. Pairs that show one color and hide different ones: (1,3), (2,3), (4,5), (4,6).

Pairs that show alike but hide different colors

Grades 2–3 P1 · first route

Idea. Ticket n shows face n and hides the other face of its card. Red-showing tickets 1, 2, 3 hide red, red, blue and blue-showing 4, 5, 6 hide red, blue, blue, so the pairs are 1,3, 2,3, 4,5, 4,6.

Exact hidden-color chances for each clue

Grades 4–5 P1 · first route

Idea. Red shows on tickets 1, 2, 3, which hide red, red, blue: P(hidden red | red showing) = 2/3. By symmetry P(hidden blue | blue showing) = 2/3.

Red showing: removals that make a tie

K–1 P2 · first route

Idea. Red shows on tickets 1, 2, 3: hidden red has two ways, blue one. Removing ticket 1 or ticket 2 makes a tie.

Blue showing: removals that make a tie

K–1 P3 · first route

Idea. Blue shows on tickets 4, 5, 6, which hide red, blue, blue. Removing ticket 5 or ticket 6 makes a tie.

The better guess after a red clue

Grades 2–3 P2 · first route

Idea. Three equally likely tickets give a red clue; two of them hide red, so red is the better guess, 2 ways out of 3.

The better guess after a blue clue

Grades 2–3 P3 · first route

Idea. Tickets 4, 5, 6 give a blue clue and hide red, blue, blue: blue is the better guess, 2 ways out of 3.

A friend's one-card-each argument

Grades 4–5 P2 · first route

Idea. Equally likely outcomes are the six faces, not the two red-faced cards: the red/red card has two red faces, so it produces red clues twice as often.

One fact about the covered number

Grades 4–5 P3 · first route

Idea. After a red clue faces 1, 2, 3 survive. The number decides the hidden color; odd leaves faces 1 and 3, a tie; even leaves face 2, red for certain; 3 or less removes nothing, so red stays 2 to 1.

Add one ticket to the two-ticket chooser

K–1 P4 · return route

Idea. Tickets 1 and 3 tie red against blue. Adding 2 makes hidden red more likely; adding 4, 5 or 6 adds no red-showing round, so the tie stays.

Three tickets where red always hides blue

K–1 P5 · return route

Idea. Only ticket 3 shows red over blue, so the cup must hold 3 and no 1 or 2: 3,4,5, 3,4,6, 3,5,6.

Compare the two-ticket chooser with all six tickets

Grades 2–3 P4 · Grades 4–5 P4 · return route · wording differs by band

Idea. The two-ticket chooser gives hidden red 1/2; drawing from all six and keeping red rounds gives 2/3. The same clue can mean different things under different selection rules.

Can six rounds tell the choosers apart?

Grades 4–5 P5 · return route

Idea. Any sequence of six hidden colors has positive chance under both rules, $(2/3)^k(1/3)^{6-k}$ and $(1/2)^6$, so six rounds cannot prove a difference.

Remove one card

K–1 P6 · Grades 2–3 P5 · return route · wording differs by band

Idea. Removing the blue/blue card leaves red clues from 1, 2, 3 (hidden red 2/3); removing red/blue leaves only red/red (red hides red for certain); removing red/red leaves only red/blue (red hides blue for certain). No single removal makes a tie.

A two-ticket cup with a fair red clue

K–1 P7 · Grades 2–3 P6 · return route · wording differs by band

Idea. A red clue ties exactly when the cup holds one of tickets 1 and 2 and ticket 3: 1,3 and 2,3.

Every subset with a fair red clue

Grades 4–5 P6 · return route

Idea. A red clue is fair exactly when the cup holds ticket 3 and exactly one of tickets 1 and 2; tickets 4, 5, 6 are free: $2 \times 2^3 = 16$ subsets.

Whole cards cannot make the red clue fair

Grades 4–5 P7 · return route

Idea. Whole cards keep tickets 1 and 2 together, so hidden red counts 0 or 2 while hidden blue counts 0 or 1: never a positive tie.